

After the last Ozempic shot, several clocks start ticking

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Stopping Ozempic does not flip one switch. The drug fades over weeks; appetite, weight, and blood tests can change over months; and the best evidence describes an average return of treated biology, not a guaranteed fate for one body.

Rubino's trial investigators made the moment unusually clean. After a run-in, several hundred adults who had reached the obesity dose were randomized: two-thirds kept taking semaglutide, while the rest received placebo; both groups kept lifestyle support.¹ From that fork, the people switched off semaglutide gained 6.9% of their randomization weight; those who stayed on it lost another 7.9%.¹

That experiment supplies the clearest answer—and its biggest trap. Ozempic and Wegovy contain semaglutide, but the strongest withdrawal trials used the higher obesity dose associated with Wegovy in people without diabetes.^{1,2} Those trials do not directly tell us what happens after lower Ozempic doses or in diabetes.^{1,2} The active drug is the same; the tested dose, starting point, and stakes are not.^{1,2} What happens after the last injection is best understood as a sequence of clocks, each running at a different speed.

01 The drug clock runs for weeks

The final injection keeps working after the pen is put away.^{3,4} Overgaard's team modeled semaglutide in blood after oral, subcutaneous, and intravenous dosing; Yang's group later gathered the human pharmacokinetic studies.^{3,4} Both put its plasma half-life near one week, with measurable exposure lingering for several weeks.^{3,4} Biology gets a fade, not a cliff.^{3,4}

As the exposure ebbs, some unwanted effects can recede too.⁵ Wharton and colleagues found that nausea, diarrhoea, vomiting, and constipation clustered during active treatment and dose escalation.⁵ In the withdrawal phase, new gut events were less common after the switch to placebo.⁵ The evidence therefore looks less like a classic acute withdrawal syndrome than the gradual loss of a drug's ongoing effects.^{1,5,2} The trials did not, however, record every symptom from every person's final dose.²

The staggered sequence is mapped in Figure 1: exposure fades first, while the outcomes it had been holding change on their own clocks.^{3,6,2,7}

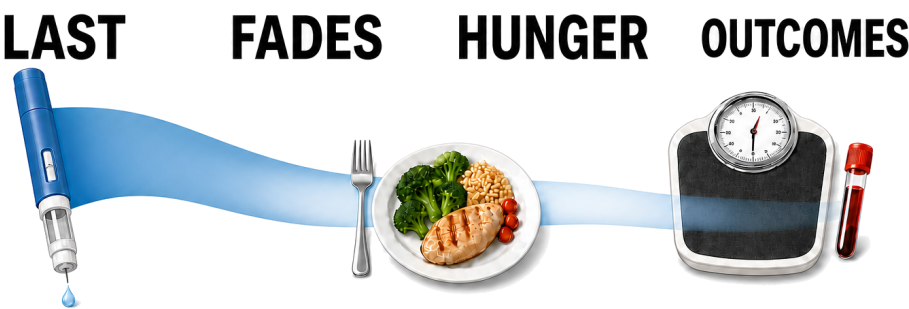


Figure 1. One injection ends; its effects unwind on different clocks. Follow the blue ribbon from the final dose: semaglutide exposure falls over weeks, appetite can re-emerge as that signal thins, and average weight and metabolic changes unfold later. The spacing is schematic, not a schedule for one person.^{3,6,2,7}

02 Hunger can return before the scale moves

What disappears with the drug? While semaglutide is active, the GLP-1 receptor agonist, a medicine that mimics a gut-hormone signal involved in appetite and glucose control, tamped down appetite and food intake across three controlled feeding studies.^{6,8,9} In the measured meals, people ate roughly one-quarter to one-third less.^{6,8}

The stomach story is subtler. Two controlled studies found no durable delay in gastric emptying by their indirect paracetamol test.^{8,9} Appetite, cravings, and control of eating carry the better-supported explanation.^{6,8,9} When the pharmacological signal fades, those effects can fade with it.^{10,11}

But here the evidence jumps a missing stair. Researchers have measured hunger carefully while semaglutide is active.^{6,8,9} They have rarely repeated those measurements

after it stops.^{10,11} Reviews propose returning appetite, counter-regulatory hormones, and energy-balance adaptations as drivers.^{10,11} The week-by-week return of hunger remains largely uncharted.^{10,11}

03 By months, the average weight curve points uphill

Rubino's randomized split shows that lifestyle support did not stop the average curve from turning.¹ Wilding's team watched selected trial completers for a full year after semaglutide and structured counseling both ended.² They had lost about one-sixth of their starting weight, then regained roughly two-thirds of that loss; nearly half still retained a clinically meaningful loss.² Regain was common, but complete erasure was not.²

Panel A of Figure 2 shows the randomized fork; panel B follows the longer out-and-back path.

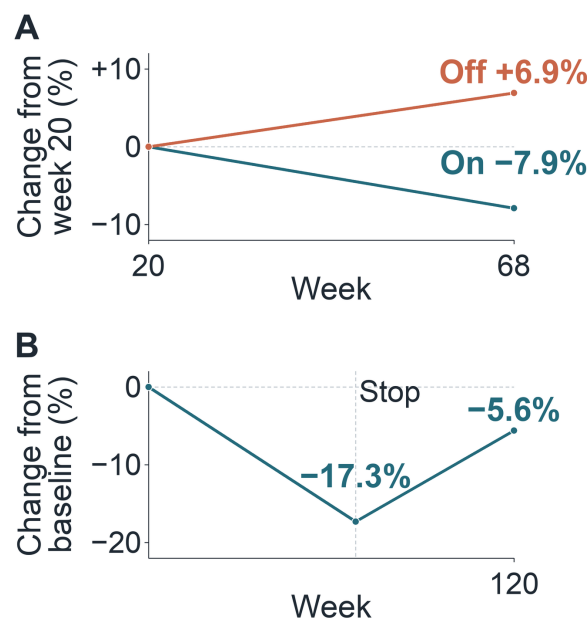


Figure 2. The group average turns upward after semaglutide stops. In panel A, the x-axis is STEP 4 trial week 20 to 68 and the y-axis is mean percentage change from week-20 weight; the direct labels mark the continued and withdrawn endpoints. The table below reports their contrast and interval. In panel B, the x-axis is STEP 1 week 0 to 120 and the y-axis is mean percentage change from pretreatment baseline. Points are group means, not individual tracks.^{1,2,12,13}

The pooled literature agrees on the direction and quarrels about the amount.^{12,7,14,13} Berg's and Tzang's teams both found regain after incretin drugs stopped, with sharp differences among drugs, populations, and trials.^{12,7} West's

broader review and Budini's nonlinear model likewise traced an average path back upward.^{14,13} Models that project a plateau go beyond direct data, which stop at roughly a year.^{13,14}

EVIDENCE	STOP	AVERAGE	LIMIT
STEP 4 ¹	803; 2.4 mg continued or stopped	Weeks 20–68: –7.9% vs +6.9%; –14.8-point difference (95% <u>confidence interval</u> , plausible range –16.0 to –13.5)	Not Ozempic doses; run-in intolerance
STEP 1 extension ²	327; 2.4 mg stopped	–17.3%, then +11.6 points: –5.6% net after a year	No longer-term outcomes; no diabetes-specific inference; no individual prediction
HIV cohort ¹⁵	49; 1 mg stopped	–7.8 then +2.9 kg in 24 weeks	No control; limited generalisability
PCOS cohort ¹⁶	25; 1 mg stopped; metformin	101→92→95 kg over two years	Metformin or selection

04 Blood tests follow, but on another clock

Weight is only the visible dial. In the STEP 1 extension, blood pressure returned to baseline and improvements in glucose, inflammation, and several lipids narrowed as weight came back, although some advantages remained.² Tzang's randomized-trial synthesis found the same overall direction, but not a synchronized march.⁷ In obesity studies, systolic pressure and HbA1c—a measure of average blood sugar over recent months—rose on average, while HDL cholesterol and diastolic pressure did not clearly worsen.⁷

Diabetes changes the balance. Across randomized diabetes trials, average weight regain was smaller than in obesity trials, while average HbA1c worsened more.⁷ The fasting-glucose estimate was too imprecise to rule out little change.⁷ An active-treatment trial also found that far more participants with prediabetes returned to normal-range measures on semaglutide than on placebo.¹⁷ Losing that control can matter even when the scale moves modestly.^{7,18}

Could stopping also change heart attacks or strokes? Xie's team emulated a target trial in Veterans Affairs records and found progressively higher cardiovascular-event risk with longer GLP-1 treatment gaps versus continuous use.¹⁹ That pattern is concerning, but it remains observational.¹⁹ Prescriptions are not ingestion, the population was mostly veterans with diabetes, and unmeasured differences may remain.¹⁹ Current reviews still call hard outcomes after discontinuation an open question.¹⁸

05 Then the neat average breaks apart

Gasoyan and colleagues supplied the plot twist. In a large health-record cohort using injectable semaglutide or tirzepatide, people who stopped early lost less weight by one year than those who continued.²⁰ Yet the recorded weights after discontinuation looked surprisingly stable.²⁰ That does not overturn the randomized trials.^{1,2,20} Most patients used

lower maintenance doses, clinical weights were sparse, prescription gaps could reflect shortages, and diet, another medicine, or a restart could happen offstage.²⁰

Smaller prospective studies widen the range.^{15,16} Erlandson's lower-dose cohort regained some, not all, of its loss over the following months.¹⁵ In Jensterle's small PCOS cohort, participants continued metformin and retained about two-thirds of the semaglutide loss two years later.¹⁶ These studies cannot prove that a lower dose is easier to stop.^{15,16} They do show that “everyone returns to baseline” is too strong.^{15,16}

The scale hides another unknown. During active treatment, dual-energy X-ray absorptiometry, an imaging method that separates fat from lean tissue, showed semaglutide can reduce both.^{21,22} Reviews warn that “lean mass” also includes water and organs and need not mean lost muscle function.^{23,24} Almost no study has repeated those scans after withdrawal.^{22,23,24} The returning kilograms do not arrive with labels.

This unfinished question reaches ordinary care.^{25,26} In a US cohort, stopping within a year was common—especially without diabetes—and many people later restarted.²⁵ Swedish diabetes registers likewise showed that refill gaps often ended in reinitiation.²⁶ “Stopped” is often a chapter, not an ending.^{25,26}

Why can a clean randomized mean and steadier clinic records both be true? The studies ask different questions. Withdrawal trials fix the switch and follow selected groups.^{1,2} Health records mix doses, shortages, restarts, replacement medicines, sparse weigh-ins, and many reasons for stopping.^{20,25,26} Smaller cohorts add still more selected populations.^{15,16} Together, these designs can describe a group direction, but none can assign one person's path.^{2,7,20}

Figure 3 separates what the evidence can predict from what it cannot.

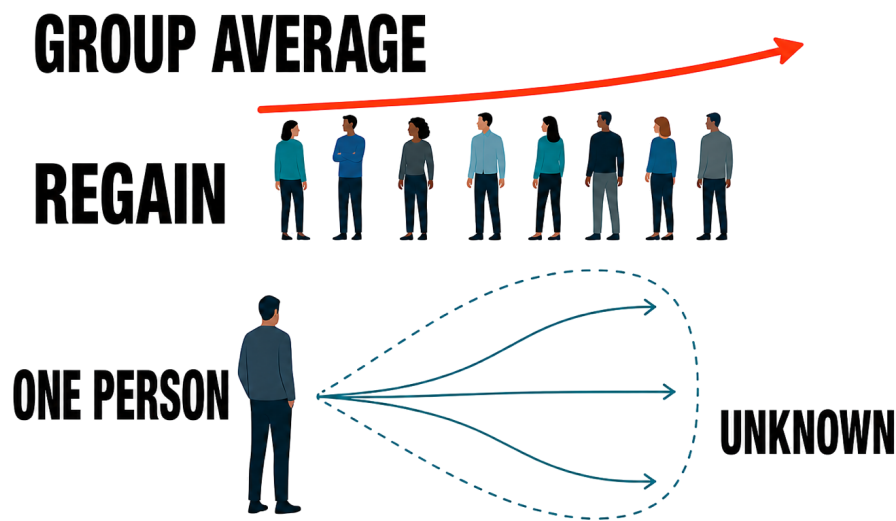


Figure 3. A reliable average still makes a poor personal forecast. The crowd and coral arrow carry the randomized group finding: withdrawal usually bends the average toward regain. The lone person faces three deliberately unranked paths, while the dashed boundary marks the missing rule that would assign one person to one course.^{1,20,15,16}

06 The off-ramp remains an experiment

One randomized clue comes from a related drug.²⁷ Jensen's team followed people for a year after liraglutide, supervised exercise, both, or placebo ended.²⁷ Those previously assigned exercise plus liraglutide maintained more weight loss than those previously assigned liraglutide alone; prior exercise also beat prior liraglutide after treatment stopped.²⁷ This was a demanding supervised program and a different drug.²⁷ It does not show that ordinary advice, or a few gym visits, cancels semaglutide rebound.^{27,14}

Broad reviews have not found a clear average shield from routine behavioral support after medication cessation.¹⁴ No high-quality randomized semaglutide trial has shown that tapering, intermittent dosing, or lower-dose maintenance beats abrupt stopping.^{10,28,29} Clinical reviews discuss those routes because the gap is real, not because the answer is settled.^{10,28,29} Continued treatment with the related dual agonist tirzepatide preserves far more weight loss than withdrawal.³⁰ Its metabolic rebound tracks the amount regained, but that reinforces chronic-treatment biology rather than solving the off-ramp.³¹

07 The last shot is a starting gun, not a forecast

Return to Rubino's trial. The last injection did not trigger one event.^{3,1,2} Semaglutide exposure ebbed first.^{3,4} Its appetite effects could fade next.^{6,8,9,10} Weight and metabolic markers followed on slower clocks.^{1,2,7} The person-level paths spread beyond the bold trial mean.^{2,20}

The evidence lands firmly on direction and softly on destiny.^{1,2,7,20} After semaglutide stops, gastrointestinal effects often ease.⁵ Appetite effects can fade.^{6,8,9,10} Average weight commonly rises.^{1,2} Gains in glucose, blood pressure, and lipids can shrink.^{2,7} Many people retain some weight loss, and selected lower-dose cohorts sometimes retain more.^{2,15,16} At Ozempic doses and in diabetes, the evidence is thinner, and loss of glucose control may matter more than the kilograms.^{7,18}

A trial that randomizes semaglutide users to abrupt stopping, tapering, lower-dose maintenance, or continuation—while repeatedly measuring appetite, drug concentration, body composition, glucose, and symptoms for more than a year—would finally synchronize the clocks.^{10,28,11} Until then, the last shot tells you when the question begins, not where one body will finish.

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